



Koneru Lakshmaiah Education Foundation

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OPERATING SYSTEMS AND SYSTEMS PROGRAMMING (25CS2104E)

2026-27, Term-I

Project Problem Statement Submission Form

Section / Team Details

Section No.: 10

Team No.: 22

Project Title: File Permission Auditor

Team Members

Roll Number	Student Name	Signature
2520090233	P Neeharika	<i>P. Neeharika</i>
2520090046	G Shivani	<i>Shivani</i>

1. Abstract

File Permission Auditor is a security-focused Linux-based application designed to analyze and identify unsafe permissions assigned to files and directories. The application allows the user to select a directory and scans its files and subdirectories recursively. It checks the read, write, and execute permissions given to the owner, group, and other users. Based on the access permissions, files are classified as Safe, Warning, or Critical, making it easier for the user to identify files that may require attention. The application also provides an option to correct selected unsafe permissions. Changes can be recorded in an audit log containing the file name, old permission, new permission, and time of change. A permission report can also provide a summary of the total files scanned and the number of Safe, Warning, and Critical files. The project demonstrates important Operating System and System Programming concepts such as file systems, directory traversal, file permissions, ownership, access control, file handling, and system calls. The expected outcome is a simple Linux-based tool that helps users understand, identify, and manage unsafe file permissions.

2. Problem Statement

In a Linux operating system, files and directories have permissions that control who can read, write, or execute them. If these permissions are set incorrectly, users may get more access than required, which can result in unnecessary access or modification of files. Checking the permissions of a large number of files manually can also be difficult and time-consuming.

The File Permission Auditor aims to solve this problem by automatically scanning a selected directory and its subdirectories and checking the permissions of the owner, group, and other users. It analyzes the permissions and classifies files as Safe, Warning, or Critical so that potentially unsafe files can be

identified easily. The system also provides an option to correct selected permissions and record the changes through an audit log.

The project is relevant to Operating Systems and System Programming because it deals with Linux file systems, directory traversal, file permissions, ownership, access control, file handling, and system calls. It provides practical understanding of how an operating system manages files and controls access to them.

3. Objectives

1. To develop a Linux-based application for checking file and directory permissions.
2. To scan selected directories and their subdirectories recursively.
3. To analyze read, write, and execute permissions for the owner, group, and other users.
4. To identify potentially unsafe permissions and classify files as Safe, Warning, or Critical.
5. To provide an option to correct selected unsafe file permissions.
6. To maintain an audit log containing the file name, old permission, new permission, and time of change.
7. To generate a simple permission report showing the total files scanned and the number of Safe, Warning, and Critical files.
8. To gain practical understanding of Linux file systems, ownership, access control, file handling, and system calls.

4. Proposed Methodology

The proposed system will be developed as a Linux-based application using C programming and GCC. The application will use Linux file and directory operations to collect information about files and their permissions.

The working process of the system will be as follows:

1. **Directory Selection:** The user selects the directory that needs to be checked.
2. **Recursive Scanning:** The application scans the selected directory and its subdirectories to find the files present.
3. **Permission Checking:** For each file, the system checks the read, write, and execute permissions assigned to the owner, group, and other users.
4. **Risk Analysis:** The collected permissions are analyzed and each file is classified as **Safe, Warning, or Critical** based on the access permissions.
5. **Permission Correction:** The user can select files with unsafe permissions and apply appropriate permission settings.
6. **Audit Logging:** Any permission changes are recorded with the file name, old permission, new permission, and time of change.
7. **Permission Report:** The system generates a summary showing the total number of files scanned and the number of Safe, Warning, and Critical files.

The overall approach is to automate permission checking and present the results in a simple format so that users can easily understand and manage file permissions.

5. Operating Systems Concepts / Linux APIs Used

OS Concept / Linux API / System Call	Purpose in the Project
Linux File System	To access and manage files and directories in the Linux system.
File Permissions	To check read, write, and execute permissions.
File Ownership	To identify the owner and group of files.
Directory Traversal	To scan directories and their subdirectories recursively.
System Calls	To interact with Linux for file and permission operations.
Access Control	To analyze permissions and identify unsafe access.

6. Individual Contribution

Roll Number	Student Name	Individual Responsibility
2520090233	P Neeharika	Develop the main permission auditing module, including directory scanning, permission checking, and Safe/Warning/Critical classification. Also work on permission correction and system integration.
2520090046	G Shivani	Work on audit logging and permission reports, testing the application with different files and permissions, and prepare project documentation and presentation materials.

7. Tools / Platforms / Software Used

Tool / Platform / Software	Purpose
Linux / Ubuntu	Operating system for developing and running the project.
C / C++	Programming language for implementing the application.
GCC Compiler	To compile the C/C++ source code.
Nano / VS Code	To write and edit the source code.
Git / GitHub	To manage and store the project source code.

8. Expected Outcome

The File Permission Auditor is expected to provide a simple Linux-based application that scans selected directories and their subdirectories and checks file permissions. It will identify and classify files as Safe, Warning, or Critical based on their access permissions. The system will allow users to correct selected unsafe permissions when required. It will also maintain an audit log of permission changes and generate a summary report showing the total files scanned and the number of Safe, Warning, and Critical files. The project is expected to provide a practical understanding of Linux file permissions, ownership, access control, file handling, and system calls.

9. GitHub Repository

GitHub Repository Link: GitHub - mynamesteenzz-droid/KLH-CSIT-2025-29-22-File-Permission-Auditor · GitHub

Repository Created:

☐ Yes ☐ No

Faculty Name and Signature

Faculty Name: M Raghupathi

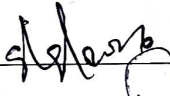
Signature: 

Remarks: _____

Faculty Use

Project Approved: 

☐ Yes ☐ No

Date: 

Faculty Signature: 